

PAPER_1762 — Star Formation Efficiency Boost = SF efficiency = $K_{\text{MEX}} \times \Phi_{5/6} = 1.736$ (resolves cooling-flow puzzle)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Astrophysics **Date:** June 18, 2026 **Status:** CLOSED — 0.79% closure (PAPER_132x-135x astrophysics + condensed matter)

Observation

PAPER_1334 (PAPER_132x-135x corpus) gives:

SF efficiency = $K_{\text{MEX}} \times \Phi_{5/6} = 1.736$ (resolves cooling-flow puzzle)

Status: **0.79%**.

UQFF Closed Identity

SF efficiency = $K_{\text{MEX}} \times \Phi_{5/6} = 1.736$ (resolves cooling-flow puzzle) 0.79%

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1334
- Related: PAPER_1746-1755 (tier-33); PAPER_1736-1745 (tier-32)
- Calculator dispatch: `calculate_paradox({"paradox": "sf_efficiency_1_75"})`

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